

LAB MST

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SECTION: 24AIT_KRG_1(A)

SUBJECT: ADBMS

DATE: 9/10/26

SOLUTION 1:

```
FROM customer c
```

```
JOIN orders o ON c.customer_id=o.customer_id
```

```
GROUP BY c.customer_id, c.customer_name, c.city
```

```
HAVING SUM(o.amount)=(
```

```
    SELECT MAX(x.total)
```

```
    FROM (
```

```
        SELECT c2.city,SUM(o2.amount) as total
```

```
        FROM customer c2
```

```
        JOIN orders o2 on c2.customer_id=o2.customer_id
```

```
        GROUP BY c2.customer_id, c2.city
```

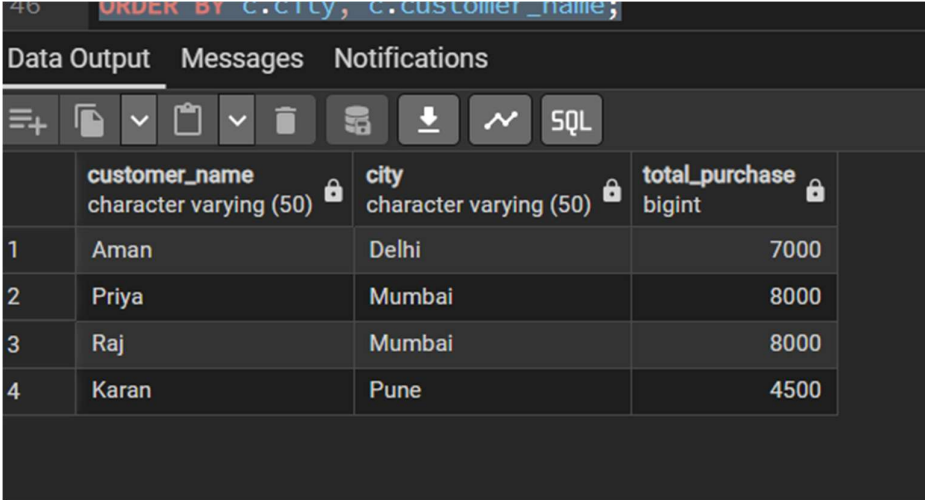
```
    ) x
```

```
    WHERE x.city=c.city
```

```
)
```

```
ORDER BY c.city, c.customer_name;
```

OUTPUT 1:



	customer_name character varying (50)	city character varying (50)	total_purchase bigint
1	Aman	Delhi	7000
2	Priya	Mumbai	8000
3	Raj	Mumbai	8000
4	Karan	Pune	4500

SOLUTION 2:

```
CREATE TABLE product_audit (  
    audit_id    SERIAL PRIMARY KEY,  
    product_id  INT,  
    product_name VARCHAR(100),  
    action      VARCHAR(20),  
    action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
CREATE OR REPLACE FUNCTION log_product_changes()  
RETURNS TRIGGER AS $$  
BEGIN  
    IF TG_OP = 'INSERT' THEN  
        INSERT INTO product_audit (product_id, product_name, action)  
        VALUES (NEW.product_id, NEW.product_name, 'PRODUCT ADDED');  
        RETURN NEW;  
    ELSIF TG_OP = 'DELETE' THEN  
        INSERT INTO product_audit (product_id, product_name, action)  
        VALUES (OLD.product_id, OLD.product_name, 'PRODUCT REMOVED');  
        RETURN OLD;  
    END IF;  
END;  
$$ LANGUAGE plpgsql;  
  
CREATE TRIGGER trg_product_audit  
AFTER INSERT OR DELETE ON product_inventory  
FOR EACH ROW  
EXECUTE FUNCTION log_product_changes();  
  
INSERT INTO product_inventory VALUES (101, 'Laptop', 65000), (102, 'Mobile', 30000);  
  
TRUNCATE product_audit RESTART IDENTITY;  
  
INSERT INTO product_inventory VALUES (103, 'Tablet', 20000);  
  
DELETE FROM product_inventory WHERE product_id = 101;
```

SELECT audit_id, product_id, product_name, action FROM product_audit;

OUTPUT 2:

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postgres/postgres@PostgreSQL 18

No limit

QueryQuery History

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EXECUTE FUNCTION log_product_changes();

INSERT INTO product_inventory VALUES (101, 'Laptop', 65000), (102, 'Mobile', 30000);

TRUNCATE product_audit RESTART IDENTITY;

INSERT INTO product_inventory VALUES (103, 'Tablet', 20000);

DELETE FROM product_inventory WHERE product_id = 101;

SELECT audit_id, product_id, product_name, action FROM product_audit;

Data OutputMessagesNotifications

Showing

	audit_id [PK] integer	product_id integer	product_name character varying (100)	action character varying (20)
1	1	103	Tablet	PRODUCT ADDED
2	2	101	Laptop	PRODUCT REMOVED